

Girijananda Chowdhury University, Guwahati

Department of Civil Engineering

B.Tech 4th Semester

Formative Assessment II

Paper Code: BCE23401T

**Subject- Transportation
Engineering**

Total Marks: 30

Time: 1 Hr

1. Define Gradient. What are the types of gradient used in Highway Engineering? 10
2. Discuss the procedure of CBR test. 10
3. A) Calculate the extra widening required for a two-lane National Highway at a horizontal curve of 300 m radius, considering a wheel base of 8 m and a design speed of 100 km/hr. 5
- B) The radius of a horizontal curve is 100 m. The design speed is 60 kmph and the design coefficient of lateral friction is 0.15. Calculate 5
 - iii. The superelevation required if full lateral friction is assumed to be developed.
 - iv. The coefficient of friction needed if no superelevation is provided.